

1 Solution — GIAM 1.2

Problem 1

1.1 Problem

Find the prime factorizations of the following integers.

- (a) 105
- (b) 414
- (c) 168
- (d) 1612
- (e) 9177

1.2 Strategy — Divisibility Rules + Trial Division up to $\sqrt{n}$

For each integer n :

1. **Repeatedly divide out the small primes:** 2 (if n is even), 3 (if the digit sum is divisible by 3), 5 (if n ends in 0 or 5). Each successful division replaces n by a smaller quotient.
2. **Then trial-divide by the remaining primes** 7, 11, 13, 17, 19, ... up to $\sqrt{n}$. If no prime up to $\sqrt{n}$ divides n , then n itself is prime.

The $\sqrt{n}$ bound — exactly the sieve observation the textbook flags in Section 1.2 — comes from: if $n = ab$ with both $a, b > \sqrt{n}$, then $ab > n$, contradiction. So at least one factor is $\leq \sqrt{n}$.

1.3 Visual Summary — The Five Factor Chains

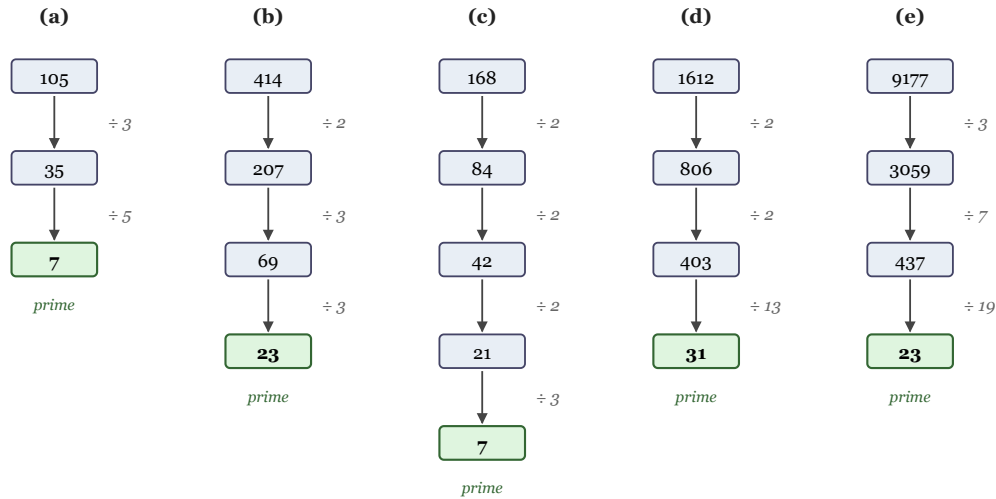


Figure 1.1: Each column shows the running quotient after dividing out one prime at a time. Composite quotients are blue; the terminal prime is green. The divisor on each arrow is the smallest prime dividing the running quotient.

1.4 Part (a) — $105 = 3 \cdot 5 \cdot 7$

- 105 ends in 5, so $5 \mid 105$: $105 = 5 \cdot 21$. Alternatively, digit sum $1 + 0 + 5 = 6$ is divisible by 3, so we can start with 3: $105 = 3 \cdot 35$.
- $35 = 5 \cdot 7$ — both prime.

$$105 = 3 \cdot 5 \cdot 7$$

1.5 Part (b) — $414 = 2 \cdot 3^2 \cdot 23$

- 414 is even: $414 = 2 \cdot 207$.
- 207: digit sum $2 + 0 + 7 = 9$, divisible by 9. So $207 = 9 \cdot 23 = 3^2 \cdot 23$.
- 23 is prime ($\sqrt{23} < 5$; not divisible by 2, 3).

$$414 = 2 \cdot 3^2 \cdot 23$$

1.6 Part (c) — $168 = 2^3 \cdot 3 \cdot 7$

- $168/2 = 84$.
- $84/2 = 42$.
- $42/2 = 21$.
- $21 = 3 \cdot 7$.

$$168 = 2^3 \cdot 3 \cdot 7$$

1.7 Part (d) — $1612 = 2^2 \cdot 13 \cdot 31$

- $1612/2 = 806$.
- $806/2 = 403$.
- **403**: odd; digit sum $4 + 0 + 3 = 7$ — not divisible by **3**. Doesn't end in **0** or **5**. Try primes up to $\sqrt{403} \approx 20.1$:
 - $403/7 = 57.57 \dots \times$
 - $403/11 = 36.6 \dots \times$
 - $403/13 = 31 \checkmark$
- So $403 = 13 \cdot 31$, and **31** is prime (no prime $\leq \sqrt{31} \approx 5.6$ divides it).

$$1612 = 2^2 \cdot 13 \cdot 31$$

1.8 Part (e) — $9177 = 3 \cdot 7 \cdot 19 \cdot 23$

- **9177** is odd; digit sum $9 + 1 + 7 + 7 = 24$, divisible by **3**. So $9177 = 3 \cdot 3059$.
- **3059**: digit sum **17** — not divisible by **3**. Not even. Doesn't end in **0** or **5**. Try primes:
 - $3059/7 = 437 \checkmark$
- **437**: odd, digit sum **14** (not div by 3), doesn't end in 0/5. Try primes up to $\sqrt{437} \approx 20.9$:
 - $437/7 = 62.4 \dots \times$
 - $437/11 = 39.7 \dots \times$

- $437/13 = 33.6 \dots \times$
- $437/17 = 25.7 \dots \times$
- $437/19 = 23 \checkmark$
- 23 is prime.

$$9177 = 3 \cdot 7 \cdot 19 \cdot 23$$

1.9 Summary

Part	Integer	Prime factorization	Distinct primes
(a)	105	$3 \cdot 5 \cdot 7$	3, 5, 7
(b)	414	$2 \cdot 3^2 \cdot 23$	2, 3, 23
(c)	168	$2^3 \cdot 3 \cdot 7$	2, 3, 7
(d)	1612	$2^2 \cdot 13 \cdot 31$	2, 13, 31
(e)	9177	$3 \cdot 7 \cdot 19 \cdot 23$	3, 7, 19, 23

Table 1.1

1.10 Remark — The Fundamental Theorem of Arithmetic

Each of the five factorizations is *unique*, up to the order of factors. This is the **Fundamental Theorem of Arithmetic**:

Every integer $n \geq 2$ has a unique prime factorization, up to reordering of the primes.

Existence comes from repeated division-by-a-prime-factor (Euclid's argument): if $n > 1$ is composite, its smallest divisor > 1 is a prime, and the quotient is strictly smaller — induct.

Uniqueness comes from Euclid’s lemma: *if a prime p divides ab then $p \mid a$ or $p \mid b$.* This is the alternative definition of “prime” the textbook flags in Section 1.2 — and it is exactly the property that forces factorizations to match. (Without Euclid’s lemma, factorizations can branch — and they do, in some number systems like $\mathbb{Z}[\sqrt{-5}]$, where $6 = 2 \cdot 3 = (1 + \sqrt{-5})(1 - \sqrt{-5})$ are two genuinely distinct prime factorizations.)

1.11 Remark — The $\sqrt{n}$ Bound, Made Concrete

The textbook’s sieve observation: “to find all primes less than p^2 , we need only use the primes up to (but not including) p .” Applied to factorization: *to factor n , we need only trial-divide by primes up to $\lfloor \sqrt{n} \rfloor$.*

Concretely, for each integer in this problem:

n	$\sqrt{n}$	Primes to try $\leq \sqrt{n}$	# of primes
105	≈ 10.2	2, 3, 5, 7	4
414	≈ 20.3	2, 3, 5, 7, 11, 13, 17, 19	8
168	≈ 12.9	2, 3, 5, 7, 11	5
1612	≈ 40.1	2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37	12
9177	≈ 95.8	2, 3, 5, 7, $\dots$, 89 (24 primes)	24

Table 1.2

In practice you stop *much* earlier — as soon as you’ve stripped enough small primes that the running quotient is small enough to recognise as a known prime.

1.12 Remark — Divisibility Rules in Use

The rules used silently above:

Prime p	Rule for $p \mid n$	Example
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2	last digit is even	$414 \rightarrow 4$ — even ✓
3	digit sum divisible by 3	$105 \rightarrow 1 + 0 + 5 = 6$ ✓
5	last digit is 0 or 5	$105 \rightarrow 5$ ✓
9	digit sum divisible by 9	$207 \rightarrow 2 + 0 + 7 = 9$ ✓
11	alternating digit sum divisible by 11	$1331 \rightarrow 1 - 3 + 3 - 1 = 0$ ✓

Table 1.3

There are similar tricks for **7, 13, 17** but they’re less elegant; usually quicker to just trial-divide. The rules for **2, 3, 5, 11** are by far the most useful in practice.

1.13 Remark — Why (e) Is the Trickiest

Item (e), **9177**, is the hardest because *both* large prime factors are unfamiliar — neither **19** nor **23** is among the “first ten” primes that most people have memorised. The work to identify $19 \cdot 23 = 437$ requires checking every prime up to $\sqrt{437} \approx 20.9$, which is essentially trial division through every prime through **19**.

A useful mental shortcut: once the running quotient is below a few thousand, *square-root every step* to narrow the search. For **437**: $\sqrt{437} \approx 21$, so I need to check only **7, 11, 13, 17, 19** (having already ruled out **2, 3, 5**).

1.14 Remark — Distinct Primes vs Repeated Primes

The “shape” of a factorization (which primes appear with what multiplicity) is informative on its own. The integers above sort into two groups:

- **Square-free** (no repeated prime): **105, 1612, 9177**. These have factorization $p_1 p_2 \cdots p_k$ with distinct primes.
- **Has a repeated prime**: $414 = 2 \cdot 3^2 \cdot 23$ (the **3** repeats), $168 = 2^3 \cdot 3 \cdot 7$ (the **2** repeats).

A number is square-free iff n is not divisible by any *perfect square* > 1 . Square-free numbers play a special role in number theory (for instance, the Möbius function $\mu(n)$ is

zero exactly on the non-square-free integers).

1.15 Remark — Connection to Cryptography

Prime factorization sits at the heart of **RSA encryption**: the security of RSA depends on the assumption that *factoring* a large composite is hard, while *multiplying* primes is easy. For an integer of the size we’re factoring here ($\leq 10^4$), trial division finishes in milliseconds. But for RSA-grade integers (say $\sim 10^{600}$), even the best known algorithms take astronomical time on classical computers.

The exercise in this problem is “factoring is straightforward for small n .” The exercise that secures the internet is “factoring is conjectured to be infeasible for large n .” The gap between those is the entire field of computational number theory.